

CS3490 Homework 3

Combinatory Logic and Lambda Calculus

I	=	$\lambda x.x$
K	=	$\lambda xy.x$
S	=	$\lambda xyz.xz(yz)$
B	=	$\lambda xyz.x(yz)$
C	=	$\lambda xyz.xzy$
P	=	$\lambda xyz.zxy$
ω	=	$\lambda x.xx$
Ω	=	$(\lambda x.xx)(\lambda x.xx)$

The table above lists the translation of combinators into the lambda calculus. You are to simplify the expressions below until no further beta redexes are present, or until a reduction cycle is detected. You can start by using the rules of combinatory logic that we learned during the combinatory logic lecture. When no more combinator rules can be applied, or you reach a λ , proceed with beta reduction for lambda terms. Write the solutions with pencil on separate paper(s), and submit them stapled in class on the due date. (I will hide the homework assignment from asuLearn once the deadline passes.)

1. $SCSIK\Omega$

2. $CS(\Omega I)P\omega KI$

3. $B\omega(Pxy)(KI)$

4. $K(K(Kx))\Omega\Omega\Omega$

5. $SII(BK\omega)abcd$

6. $(\lambda x.x(\lambda y.yx))I(\lambda y.yy)$

7. $S(\lambda pq.qp)(KI)I$

8. $B(\lambda xy.xyy)(\lambda xy.xyy)(\lambda z.zKI)(KI)$

9. $\omega(\lambda wx.w\omega(xI))$

10. $\omega(\lambda wx.x(\omega w(xI)))I$